RAIR and KAIR Viva Questions

Student or team names _____

Thirty questions with concise answers grounded in the implemented project and frozen results. Practice explaining the evidence and its limits in your own words.

1. What problem does this project solve?

It makes multi-hop question answering from supplied documents inspectable. The system shows retrieved evidence, intermediate knowledge, final citations and failure states, and compares two strategies under controlled inputs.

2. What does RAIR mean?

Role-Aware Iterative Retrieval. It assigns specialized responsibilities to prompts coordinated by one controller and stores accepted facts in a shared state. The roles use the same underlying model.

3. What is multi-hop question answering?

The answer requires connecting more than one fact, often across documents. In the demo the book identifies Mira Solen and a second document identifies her birthplace as Nerava.

4. Why use retrieval if the model may know the answer?

The task requires justification from the supplied sources. Retrieval selects task evidence and lets us inspect it. We do not treat model memory or a plausible uncited answer as sufficient support.

5. What is the main contribution?

A working role-based workflow with constrained source references, explicit fact admission, gap scheduling, bounded execution and a fair evaluation. We do not claim to invent decomposition or shared knowledge, because KAIR also describes a collaborative mode.

Viva questions 6 to 10

6. How does RAIR work from input to answer?

Decompose the question, plan a query for an actionable gap, retrieve with BM25, filter source sentences, extract candidate facts, verify them, update shared state and repeat. The common final synthesizer answers with citations or abstains.

7. What is the difference between a candidate and a verified fact?

A candidate is a proposed source-backed statement. A verified fact has passed this implementation's model decisions and program guards. The word verified describes admission to memory and does not guarantee truth.

8. Why did you assign IDs in software?

Generated identifiers can be malformed or refer to unavailable evidence. Runtime aliases, required candidate keys and software-assigned IDs restrict the valid reference space. Original source text and sentence indices are reconstructed and independently validated.

9. How do you prevent an unsupported fact from resolving a gap?

The verifier must provide one decision per candidate. Support, relevance, answer type and unresolved-placeholder guards control acceptance. A rejected candidate cannot resolve a gap even if the model marks its resolves flag true.

10. Why can a supported bridge fact still be useful?

A bridge fact may identify an institution or person needed for a later search without directly giving the requested city or date. Rejecting it solely because it is not the final answer type can break a valid reasoning chain. Our failure review shows this still happens.

Viva questions 11 to 15

11. How does competitive KAIR differ?

Each agent attempts the entire question with private facts and requirements. A complete valid proposal can finish early. Otherwise the proposal with the fewest unresolved requirements supplies source context for the next iteration. Private knowledge is not merged.

12. Is your KAIR an exact paper reproduction?

No. It is a documented local adaptation of competitive KAIR. It uses serial execution, bounded launches, local Qwen and supplied HotpotQA pools, and the same final synthesizer as RAIR. Those differences limit comparisons with the paper's absolute results.

13. Why give KAIR agents different seeds?

They use identical templates, so independent sampling permits diverse proposals without inventing different agent roles. A base seed plus agent offset is logged. Different seeds do not guarantee different queries.

14. What makes the comparison fair?

Both systems receive identical questions and source pools, the same model identity, preprocessing, BM25 implementation, decoding settings, common final policy and caps. Execution order is balanced and actual costs are recorded. Gold is used only after inference.

15. Do equal call caps mean equal computation?

No. RAIR averaged 12.27 LLM calls and 9657 total tokens, while KAIR averaged 9.57 calls and 11364 tokens. The limits are matched ceilings, not equal realized token budgets. We report both actual usage and correctness.

Viva questions 16 to 20

16. Why HotpotQA and what retrieval setting did you use?

HotpotQA provides multi-hop questions and sentence support annotations. We used development distractor document pools and per-question BM25 top 3. This is a controlled supplied-pool task, not full-Wikipedia or open-web retrieval.

17. How did you avoid tuning on the final test questions?

Step 5 development used 12 already exposed questions. The final manifest excluded all 125 previously exposed IDs and froze 100 further questions before inference. Step 7 only inspects saved results. Future tuning needs another untouched cohort for evaluation.

18. What are exact match and F1?

Exact match checks equality after the declared normalization. Token F1 combines token precision and recall against the gold answer. Both can penalize aliases, longer spans and digit-word differences. We preserve these metrics and show qualitative examples separately.

19. Why are 200 runs not 200 independent questions?

There are 100 questions, each repeated at two seeds. Runs sharing a question are correlated. The confidence interval resamples whole question clusters while retaining both seeds, using 10000 paired bootstrap resamples.

20. Why include abstentions and errors in the denominator?

Removing difficult runs would inflate the score. Each arm has 200 planned runs and every run contributes to answer accuracy. Coverage, abstentions, errors and costs are reported alongside correctness.

Viva questions 21 to 25

21. What is the strongest result?

Updated RAIR scored 40.5% exact match against 18.5% for the earlier RAIR core with the same current final policy. The primary gain is 22.0 percentage points with a 95% question-cluster interval of +15.0 to +29.5 points.

22. Can you say RAIR is better than KAIR?

Only as a numerical observation in this sample, not a conclusive superiority claim. RAIR scored 40.5% and KAIR 36.0%. The RAIR-minus-KAIR interval is -3.0 to +12.0 points, which includes zero. More independent questions would improve precision.

23. Why not compare Step 4 RAIR directly with Step 5 RAIR?

They used different question cohorts and final policies. The Step 5 earlier-core control runs on the same fresh cohort with the same current final policy. That contrast estimates the bundled RAIR core changes rather than a mixture of cohort and policy effects.

24. What failures remain?

RAIR has 61 abstentions and 58 answered nonexact results out of 200. Inspected traces show incomplete decomposition, rejected bridge or negative evidence, missed entities, verifier false positives and incomplete final reasoning. Some nonexact scores are wording differences.

25. What does zero invalid citations prove?

It proves the recorded source references point to available original sentences with the expected text. It does not prove those sentences semantically support the generated answer or that every required reasoning link was cited.

Viva questions 26 to 30

26. Why did the Grant Nicholas question fail?

RAIR rejected Welsh nationality as irrelevant to deciding whether he was American. Only Danny Shirley's American nationality reached memory, and the final answer incorrectly said yes for both people. Negative evidence and final completeness need stronger development tests.

27. What were KAIR's remaining errors?

There were three whole-question NoValidProposal errors and 25 agent error events. Duplicate selections, malformed source IDs, unavailable sentence selections, empty-knowledge completion checks and truncated outputs caused agent failures. Some questions survived one agent's failure.

28. What exactly does the dashboard run?

The actual local controllers through Ollama. It shows citations, gaps or agent proposals, budgets, raw calls and a JSON download. Editable custom inputs are unscored. The recorded Step 5 benchmark table is separate and cannot be changed by a live demo.

29. Does passing 207 tests mean the answers are 100 percent correct?

No. Tests verify defined software behavior, parsers, budgets, state handling and dashboard contracts. Research answer correctness is measured separately by the benchmark. Both software tests and real-model demonstrations are necessary, and neither guarantees all future answers.

30. What should be improved next?

Develop better bridge and negative-evidence handling, relation-complete decomposition, multi-entity final checks and constrained KAIR source selections on exposed cases. Then freeze a new untouched evaluation. Additional models, larger samples, semantic review and component ablations would test generality.